

Homework 5 (due: Sept. 15)

Fall 2026

Review of Lesson 5: Security Market Line

H 5-1 You are given the following information about stock *A*, stock *B*, and the market overall:

Asset	Required Return	Volatility
Stock <i>A</i>	0.03	0.50
Stock <i>B</i>	?	0.35
Market	0.06	0.25

You also know that the correlation between the returns of stock *A* and the market is -0.25 and the correlation of stock *B* with the market is 0.30 .

- (a) If the CAPM holds, calculate the required return for stock *B*.
- (b) What is the beta of a portfolio that contains \$1,500 of stock *A* and \$4,500 of stock *B*?

H 5-2 Suppose the market consists of the following four stocks:

Stock	Number of Shares	Price per Share
<i>A</i>	10,000	27
<i>B</i>	5,000	75
<i>C</i>	2,000	326
<i>D</i>	2,000	12

You own a market portfolio that includes 50 shares of stock *A*. Determine the number of shares of stock *B* in your portfolio.